Appendix I: Detailed Explanation of B2B Chuan Guang [Chwan · Gwong] (Light Transmission) Implementation Scenario in the Manufacturing Industry

This appendix aims to concretely present the implementation scenarios of core mechanisms such as "enterprise Chuan Guang Shi [Chwan · Gwong · Shrr] (light bearers)", "value protocol chains" and "three-tier gradient services" in Chapter 7 of the main text. Taking the manufacturing industry as a typical case, it dissects the complete link, role division and practical logic of B2B Chuan Guang [Chwan · Gwong] (light transmission) among manufacturing enterprises, to help readers understand the application value of the QNLOO ecosystem in the real economy, and echo the core positioning of "virtual-real integration and industrial empowerment" in the main text. All contents in this appendix are based on the underlying rules of the entire book (property rights protection, gradient profit sharing, targeted backfeed, layered computing power, etc.), without adding any core mechanisms, and only serve as scenario-based extensions.

1. Core Positioning of the Appendix

Chapter 7 of the main text clarifies that "value creation-oriented enterprises can become 'enterprise Chuan Guang Shi [Chwan · Gwong · Shrr] (light bearers)' of the ecosystem and output value externally through value protocols". Focusing on the manufacturing scenario, this appendix answers three core questions: *How do enterprise Chuan Guang Shi transmit light to other enterprises*, *How does the value protocol chain land in the industry* and *How do all ecological roles collaborate to empower industrial upgrading*, so as to realize the landing connection from theoretical mechanisms to realistic scenarios.

2. Core Essence of B2B Chuan Guang [Chwan · Gwong] (Light Transmission) in Manufacturing (Connecting to Chapter 7 of the Main Text)

The main text proposes that "Chuan Guang from individual to individual is personal cognitive growth; Chuan Guang from enterprise to enterprise is the digital reconstruction of industrial-level production relations". Specifically for the manufacturing industry, the core logic is as follows:

Leading manufacturing enterprises (enterprise Chuan Guang Shi [Chwan · Gwong · Shrr]) sort out their accumulated production processes, quality control, equipment operation and maintenance, supply chain governance, safety specifications, and cost reduction & efficiency improvement methodologies into enterprise private knowledge bases, encapsulate them into standardized manufacturing value protocols, and rely on industrial-grade sovereign AI and hardware to empower upstream and downstream small and medium-sized manufacturing enterprises and enterprises in industrial clusters through three-tier gradient services. This realizes cognitive transmission, standard unification, resource collaboration and all-region upgrading within the industry, and ultimately achieves the ecological goal of *leading enterprises driving small and medium-sized ones, and all-region industrial win-win.*

Core Correspondence (Connecting to the Main Text)

- Chuan Guang [Chwan · Gwong] subject: corresponds to "enterprise Chuan Guang Shi [Chwan · Gwong · Shrr]" in the main text (leading manufacturing enterprises, specialized and sophisticated enterprises);
- Chuan Guang [Chwan · Gwong] carrier: corresponds to "value protocol + enterprise sovereign AI" in the main text (manufacturing-exclusive value protocol + industrial-grade sovereign AI);

- Chuan Guang [Chwan · Gwong] mode: corresponds to "three-tier gradient service" in the main text (scalable brightness transmission + in-depth warmth empowerment);
- Ecological support: corresponds to "nine ecological roles" in the main text (exclusive division of labor adapted to manufacturing scenarios).

3. Manufacturing Value Protocol Chain: Complete Composition Structure (Reusing Nine Ecological Roles, Industry-Exclusive Adaptation)

It fully reuses the logics of "six supporting necessity markets" and "nine ecological roles" in Chapter 7 of the main text, and carries out exclusive adaptation combined with manufacturing scenarios. The role division, property right rules and profit models are completely consistent with the main text.

3.1 Enterprise Chuan Guang [Chwan · Gwong] Side (Leading Manufacturing Enterprises · Enterprise Chuan Guang Shi)

- Enterprise private knowledge base (core private assets): production process parameters, welding/stamping/casting standards, quality inspection specifications, equipment fault databases, upstream and downstream supply chain resources, safety control processes, energy consumption optimization schemes (strictly following the property right rules of the main text: hardware local encrypted storage, and confidential processes and core data shall never flow out);
- Manufacturing value protocol (open and inclusive framework): structured industrial guidance processes fitting into segmented manufacturing scenarios, such as *Value Protocol for Stamping Part Production Quality Control*, *Value Protocol for Auto Parts Supply Chain Matching*, *Value Protocol for Chemical Safety*

Production Processes, Value Protocol for Machining Equipment Operation and Maintenance;

- Enterprise sovereign AI + industrial hardware: training and inference all-in-one machines, industrial AI accelerator cards, edge computing devices adapted to manufacturing scenarios, deployed locally in enterprises to run core production privacy data; cloud bases (open source/closed source) run non-privacy businesses such as general supply chain matching and order collaboration (following the "cloud-local computing power layering" rule of the main text);
- Human Chuan Guang [Chwan · Gwong] team: enterprise production engineers, process experts, supply chain managers, corresponding to the third-tier human in-depth service in the "three-tier services" of the main text, responsible for offline empowerment and customized solution implementation.

3.2 Underlying Infrastructure Roles (Adapted to Industrial Scenarios, Completely Consistent with the Main Text)

- Training service providers: specialized in providing industrial large model fine-tuning, manufacturing value protocol binding training, and production logic actor development services, providing customized adaptation for leading enterprises (Chuan Guang Shi [Chwan · Gwong · Shrr] / light transmitters) and small and medium-sized enterprises (Shou Guang Fang [Show · Gwong · Fahng] / light receivers);
- Hardware manufacturers: focusing on industrial-grade sovereign AI hardware, edge computing devices, and production line AI terminals, included in the ecological profit sharing system to undertake the rigid demand for manufacturing hardware;
- Open source/closed source base manufacturers: providing industrial large model bases, following the rules of the main text — open source industrial bases are free for personal non-commercial use and charge mandatory commission for

commercial use; closed source industrial bases share profits according to cloud invocation volume and deployment scale; if base manufacturers need to obtain industrial data, they shall purchase desensitized general production data from enterprises at a fee, and free acquisition is strictly prohibited;

- Protocol chain builders: building exclusive value protocol chains for industrial clusters, using the "geographic location matching" mechanism of the main text to prioritize serving nearby manufacturing clusters (such as the Yangtze River Delta, the Pearl River Delta, county-level industrial parks), facilitating on-site collaboration by human engineers;
- Rule makers: responsible for top-level governance, on-chain auditing, gradient profit sharing, and targeted backfeed rule implementation to ensure the compliance and order of the ecosystem.

3.3 Market and Access Mode (Connecting to the Open Value Protocol Market in the Main Text)

Leading enterprises (enterprise Chuan Guang Shi [Chwan · Gwong · Shrr]) release manufacturing value protocol APIs to the "open value protocol market" described in the main text, which can be pluggable into the protocol chains of various industrial clusters, and small and medium-sized enterprises (Shou Guang Fang [Show · Gwong · Fahng]) can invoke them on demand and access flexibly; sub-protocols can be undertaken by supporting enterprises of leading enterprises and specialized and sophisticated enterprises, forming an industrial Chuan Guang [Chwan · Gwong] network of *leading enterprises → sub-protocols → small and medium-sized enterprises*, and undertaking the "ecological spillover effect" mechanism in the main text.

4. B2B Chuan Guang [Chwan · Gwong] (Light Transmission): Three-Tier Gradient Service (Fully Adapted to Manufacturing B2B Scenarios, Reusing Main Text Rules)

Strictly following the "Brightness + Warmth" three-tier gradient service system of the main text, and optimizing service content combined with manufacturing production scenarios to achieve the dual goals of "large-scale empowerment + in-depth collaboration". All pricing and service logics are completely consistent with the main text.

Tier 1: Free Logic Actor Service (Inclusive Trial, Basic Industrial Cognition)

Small and medium-sized manufacturing enterprises can invoke the manufacturing value protocols of leading enterprises for free, and logic actor AI provides automatic guidance, focusing on "basic cognitive transmission", unifying basic industrial standards, and lowering the entry threshold for small and medium-sized enterprises:

- Learning basic production safety specifications (such as explosion protection for chemical enterprises, equipment safety for machinery enterprises);
- Mastering the basic process of standardized quality inspection (such as part size inspection, product qualification rate determination);
- Understanding basic energy consumption management and daily equipment maintenance methods.

Tier 2: Paid Logic Actor Service (Brightness Transmission, Large-Scale Industrial Empowerment)

Small and medium-sized enterprises pay for invocation, and AI strictly follows the process protocols of leading enterprises to standardly guide the entire production process, realizing "large-scale empowerment without human intervention", freeing up engineers of leading enterprises and improving industrial chain efficiency in batches:

- Real-time control of production parameters (such as stamping pressure, welding

temperature, chemical reaction time), and automatic correction of process deviations;
- Automatic identification of equipment abnormalities, fault prediction, and push of operation and maintenance solutions;
- Matching basic upstream and downstream supply chain resources (such as raw material suppliers, parts outsourcing manufacturers);
- Optimizing production scheduling and energy consumption distribution to reduce production costs of enterprises.

If small and medium-sized enterprises have a solid process foundation, they can complete production upgrading at this stage without additional high human service costs.

Tier 3: Logic Actor + Enterprise Human Team Service (Warmth + Industrial Resources, In-Depth Chuan Guang [Chwan · Gwong])

After the paid AI service is exhausted, it will be forcibly upgraded to human service, corresponding to the core of "warmth empowerment" in the main text. Human engineers and process experts of leading enterprises intervene to deliver three core values (core reasons for high pricing):
- In-depth process empowerment: customized production line transformation and process optimization solutions according to the current situation of small and medium-sized enterprises' production lines, to solve complex production problems that AI cannot handle;
- Industrial resource integration: connecting upstream raw material suppliers, downstream vehicle/complete machine manufacturers, equipment service providers, and industrial financial resources to help small and medium-sized enterprises open up the supply chain link;
- Industrial trust collaboration: endorsed by leading enterprises to reduce cooperation suspicion between upstream and downstream enterprises, build a synchronized industrial cooperation community, and promote the coordinated development of industrial clusters.

This stage realizes in-depth Chuan Guang [Chwan · Gwong] from enterprise to enterprise, and truly drives small and medium-sized enterprises to achieve technological upgrading, resource upgrading and competitiveness upgrading.

5. Manufacturing-Exclusive Ecological Mechanisms (Fully Reusing the Main Text, Scenario-Based Implementation)

All ecological mechanisms follow the core rules of Chapter 7 of the main text and are implemented in combination with manufacturing scenarios. No new rules are added to ensure a logical closed loop with the entire book.

5.1 Two-way Nutrient Closed Loop (Industrial-Level Iteration, Connecting to Section 7.8 of the Main Text)

- Internal: leading enterprises' own production data, process trial and error, and equipment operation and maintenance data flow back to the enterprise private knowledge base to continuously iterate their own value protocols;
- External: production data, process problems and quality feedback of small and medium-sized enterprises after using the value protocol flow back to the knowledge base of leading enterprises after desensitization, helping leading enterprises optimize value protocols; desensitized general production data is purchased by base manufacturers at a fee (following the rule of "value flow replaces data flow" in the main text);
- Intra-organization: production experience of employees of leading enterprises and small and medium-sized enterprises is automatically precipitated into the private knowledge base of respective enterprises, becoming the core nutrient for enterprise iteration.

5.2 Ecological Spillover Effect (All-Region Industrial Prosperity, Connecting to Section 7.7 of the Main Text)

When the human engineer service of leading enterprises (enterprise Chuan Guang Shi [Chwan · Gwong · Shrr]) reaches saturation, the traffic is automatically diverted to the sub-protocol cluster (its subsidiaries, cooperative specialized and sophisticated enterprises, supporting leading enterprises). The sub-protocol cluster undertakes the in-depth service needs of small and medium-sized enterprises, and shares profits with leading enterprises upward, driving the prosperity of the entire industrial cluster Chuan Guang [Chwan · Gwong] network, and realizing "leading guidance and all-region symbiosis".

5.3 Targeted Precision Backfeed (Manufacturing-Exclusive Supplementation, Connecting to Section 7.6 of the Main Text)

On-chain ecological health monitoring identifies shortcomings in the manufacturing ecosystem in real time (such as small and medium-sized enterprises being unable to afford industrial sovereign AI hardware, lacking industrial model training services, and high implementation costs). Ecological excess profit backfeed funds are subsidized in a targeted manner to solve core industrial pain points:

- Subsidies for the purchase of industrial sovereign AI hardware for small and medium-sized enterprises to lower their technical implementation threshold;
- Technical support for industrial training service providers, encouraging them to provide cost-effective training services for small and medium-sized enterprises;
- Incentives for industrial empowerment of leading enterprises, encouraging leading enterprises to increase value protocol output and expand the scope of Chuan Guang [Chwan · Gwong];
- Subsidies for the supply of manufacturing value protocols, encouraging enterprises to supplement industrial production value protocols and fill gaps in segmented scenarios.

5.4 Property Rights and Computing Power Rules (Strictly Following the Underlying Iron Rules of the Main Text)

- Privacy data does not leave the local area: core processes, production parameters and business secrets of leading enterprises are stored in local industrial sovereign AI hardware and shall never flow out;
- Paid transaction of general data: desensitized general production and supply chain data are purchased by base manufacturers at a fee, and enterprises set prices independently to ensure the value of data assets;
- Coexistence of open source and closed source dual bases: open source industrial bases charge commissions for commercial use, and closed source industrial bases share profits on the cloud, with multiple ecosystems coexisting to avoid single monopoly;
- Cloud-local computing power layering: the cloud runs general supply chain, order matching and basic data statistics; local runs core production privacy data, to realize optimal allocation of computing power and reduce resource waste.

6. Core Comparison Between Individual Chuan Guang [Chwan · Gwong] and Enterprise Chuan Guang [Chwan · Gwong] (Manufacturing) (Connecting to the Main Text, Strengthening Cognition)

To help readers clearly distinguish the two Chuan Guang modes, combined with the logic of the main text, the core commonalities and differences are sorted out to clarify the industrial value of enterprise Chuan Guang:

Comparison Dimension	Individual-to-Individual Chuan Guang [Chwan · Gwong] (C2C, Core of Main Text)	Enterprise-to-Enterprise Chuan Guang [Chwan · Gwong] (B2B · Manufacturing, This Appendix)
Chuan Guang [Chwan · Gwong] subject	Individuals (individual Chuan Guang Shi [Chwan · Gwong · Shrr])	Leading manufacturing enterprises, specialized and sophisticated enterprises (enterprise Chuan Guang Shi [Chwan · Gwong · Shrr])
Chuan Guang [Chwan · Gwong] content	Personal experience, professional methodology, life cognition	Production processes, industrial standards, supply chain governance, industrial production relations
Service scale	One-to-one, small scope, focusing on personal growth	Batch empowerment of industrial chains and industrial clusters, focusing on all-region industrial upgrading
Warmth carrier	Personal human communication, emotional care, cognitive companionship	Enterprise engineer teams, industrial resources, supply chain

Comparison Dimension	Individual-to-Individual Chuan Guang [Chwan · Gwong] (C2C, Core of Main Text)	Enterprise-to-Enterprise Chuan Guang [Chwan · Gwong] (B2B · Manufacturing, This Appendix)
		endorsement, offline industrial collaboration
Hardware / computing power	Personal sovereign AI terminals, lightweight hardware	Industrial-grade sovereign AI, edge computing devices, training and inference all-in-one machines
Ecological value	Personal cognitive prosperity, individual ability improvement	Real economy industrial upgrading, production relation reconstruction, national manufacturing competitiveness improvement
Core commonalities	1. Both follow the mode of "value protocol + logic actor (AI) brightness transmission + human warmth empowerment"; 2. Reuse three-tier gradient service, property right protection, gradient profit sharing, targeted backfeed,	—

Comparison Dimension	Individual-to-Individual Chuan Guang [Chwan · Gwong] (C2C, Core of Main Text)	Enterprise-to-Enterprise Chuan Guang [Chwan · Gwong] (B2B · Manufacturing, This Appendix)
	and nutrient closed loop; 3. The core goals are cognitive transmission, value creation and ecological win-win.	

7. Implementation Case (Concrete Presentation to Help Readers Understand)

Taking a leading auto parts enterprise (enterprise Chuan Guang Shi [Chwan · Gwong · Shrr]) as an example, it fully presents the practical process of B2B Chuan Guang [Chwan · Gwong] in manufacturing, which fits all mechanisms of the main text:

1. The leading enterprise accumulates its 20 years of experience in stamping process, welding quality control, parts supply chain matching, safety production specifications, etc., sorts them into an enterprise private knowledge base, encapsulates them into the *Auto Parts Production Value Protocol*, entrusts a training service provider to complete industrial large model fine-tuning and logic actor development, and deploys them on industrial-grade sovereign AI hardware;
2. The leading enterprise releases the value protocol API to the open value protocol market and accesses the Yangtze River Delta auto parts industrial cluster protocol chain;
3. Small and medium-sized parts manufacturers (Shou Guang Fang [Show · Gwong · Fahng]) in the Yangtze River Delta region access the value protocol through the

market, first use the first-tier service for free to learn basic production safety specifications and quality inspection processes;

4. According to their own needs, small and medium-sized enterprises pay to invoke the second-tier service. AI controls stamping and welding production parameters in real time, optimizes equipment operation and maintenance, matches basic supply chain resources, and realizes improved production efficiency and higher qualification rate;
5. Some small and medium-sized enterprises with weak process foundation upgrade to the third-tier service. Human engineers and experts from leading enterprises visit to customize production line transformation solutions, connect with vehicle manufacturers and high-quality raw material suppliers, and help them open up the supply chain;
6. Production data of small and medium-sized enterprises after using the value protocol flows back to leading enterprises after desensitization for optimizing value protocols; desensitized general production data is purchased by open source/closed source base manufacturers at a fee;
7. When the annual profit of the leading enterprise exceeds the threshold of 5 million yuan, 2% excess profit is automatically accrued and injected into the Ecosystem Targeted Backfeed Public Pool to subsidize the industrial hardware procurement of surrounding small and medium-sized manufacturers;
8. When the human service of the leading enterprise is saturated, traffic is diverted to its subsidiaries (sub-protocol cluster). The subsidiaries undertake in-depth services for small and medium-sized enterprises, and share profits with the leading enterprise upward, forming a complete industrial Chuan Guang [Chwan · Gwong] network of *leading enterprise → sub-protocol → small and medium-sized enterprises*, and realizing unified quality, cost reduction and efficiency increase across the entire industrial chain.

8. Appendix Summary

Through the complete dissection of the B2B Chuan Guang [Chwan · Gwong] scenario in the manufacturing industry, this appendix concretely presents the landing path of the core mechanism of the QNLOO value protocol chain in the main text in the real economy, and clarifies the role positioning of enterprise Chuan Guang Shi, the industrial application of value protocols, and the collaborative logic of all ecological roles.

As the core carrier of the real economy, the implementation of the enterprise Chuan Guang [Chwan · Gwong] mode in the manufacturing industry confirms the core advantages of the QNLOO ecosystem: *virtual-real integration*, *sovereign autonomy*, *and win-win for all roles*. It not only protects enterprise data sovereignty and guarantees legitimate profits of all roles, but also realizes large-scale transmission of industrial cognition and all-region industrial upgrading. It provides a reusable scenario template for enterprise Chuan Guang [Chwan · Gwong] in other industries (such as agriculture, service industry), and echoes the ultimate positioning of the whole book as "civilization-level production relation infrastructure".